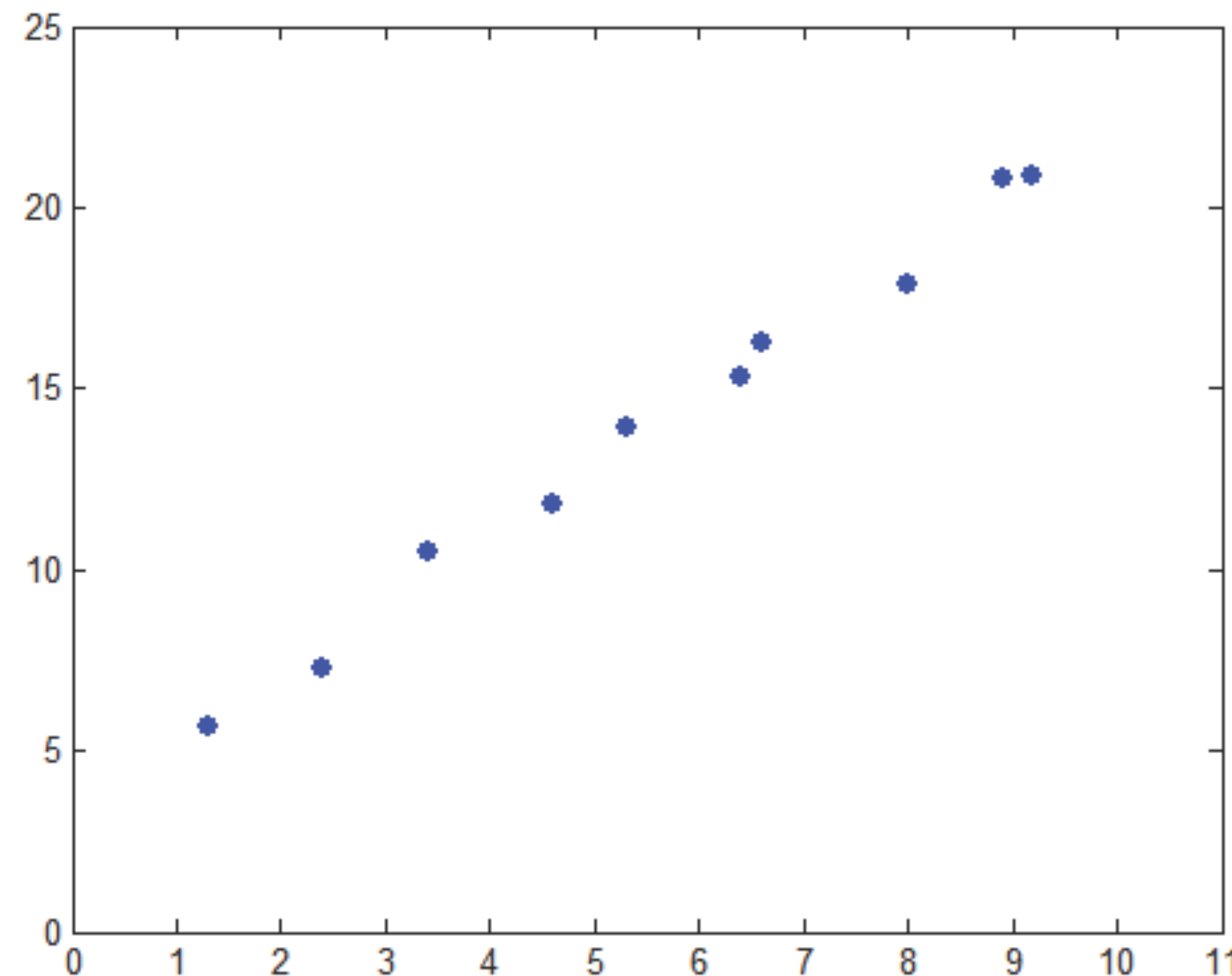


Line Fitting: Least Square Error Solutions

- ▶ Let us first consider the 'simpler' problem of fitting a line to a set of data points...

x	t
1.3	5.7
2.4	7.3
3.4	10.5
4.6	11.8
5.3	13.9
6.6	16.3
6.4	15.3
8.0	17.9
8.9	20.8
9.2	20.9



- ▶ Equation of best fit line ?

Line Fitting: Least Square Error Solutions

- ▶ Step 1: Identify the model
 - ▶ Equation of line: $y = mx + c$

Line Fitting: Least Square Error Solutions

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 - ▶ Equation of line: $y = mx + c$
- ▶ Step 2: Set up an error term which will give the goodness of every point with respect to the (unknown) model
 - ▶ Error induced by i^{th} point: $e^{(i)} = (t^{(i)} - mx^{(i)} - c)^2$

- ▶ Error for whole data:
$$E = \frac{1}{n} \sum_i^n (t^{(i)} - y^{(i)})^2$$
$$E = \frac{1}{n} \sum_i^n (t^{(i)} - mx^{(i)} - c)^2$$

Line Fitting: Least Square Error Solutions

- Understanding of Error Function

Line Fitting: Least Square Error Solutions

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- ▶ Step 3: Differentiate Error w.r.t. parameters, put equal to zero and solve for minimum point
- ▶ In other words, find the set of parameters m and c for which the error term E is minimized

Line Fitting: Least Square Error Solutions

$$E = \frac{1}{n} \sum_i (t^{(i)} - mx^{(i)} - c)^2$$

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Line Fitting: Least Square Error Solutions

$$E = \frac{1}{n} \sum_i (t^{(i)} - mx^{(i)} - c)^2$$
$$\frac{\partial E}{\partial m} = -\frac{2}{n} \sum_i (t^{(i)} - mx^{(i)} - c)x^{(i)}$$

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Line Fitting: Least Square Error Solutions

$$\frac{\partial E}{\partial m} = ?$$

$$\frac{\partial E}{\partial m} = -\frac{1}{n} \sum_i (t^{(i)} - mx^{(i)} - c)x^{(i)}$$

$$-\frac{2}{n} \sum_i (t^{(i)} - mx^{(i)} - c)x^{(i)} = 0$$

$$-\sum_i t^{(i)}x^{(i)} + m \sum_i (x^{(i)})^2 + c \sum_i x^{(i)} = 0$$

$$\sum_i m(x^{(i)})^2 + \sum_i cx^{(i)} = \sum_i t^{(i)}x^{(i)}$$

$$\begin{matrix} \left[\sum_i (x^{(i)})^2 & \sum_i x^{(i)} \right] & \begin{bmatrix} m \\ c \end{bmatrix} & = & \left[\sum_i t^{(i)}x^{(i)} \right] \\ 1 \times 2 & & 2 \times 1 & & 1 \times 1 \end{matrix}$$

Line Fitting: Least Square Error Solutions

$$\frac{\partial E}{\partial m} = ?$$

$$\frac{\partial E}{\partial m} = -\frac{1}{n} \sum_i (t^{(i)} - mx^{(i)} - c)x^{(i)}$$

$$-\frac{2}{n} \sum_i (t^{(i)} - mx^{(i)} - c)x^{(i)} = 0$$

$$-\sum_i t^{(i)}x^{(i)} + m \sum_i (x^{(i)})^2 + c \sum_i x^{(i)} = 0$$

$$\sum_i m(x^{(i)})^2 + \sum_i cx^{(i)} = \sum_i t^{(i)}x^{(i)}$$

$$\begin{bmatrix} \sum_i (x^{(i)})^2 & \sum_i x^{(i)} \end{bmatrix} \begin{bmatrix} m \\ c \end{bmatrix} = \begin{bmatrix} \sum_i t^{(i)}x^{(i)} \end{bmatrix}$$

$$1 \times 2$$

$$2 \times 1$$

$$1 \times 1$$

$$\frac{\partial E}{\partial c} = ?$$

$$\frac{\partial E}{\partial c} = -\frac{1}{n} \sum_i (t^{(i)} - mx^{(i)} + c)$$

$$-\frac{2}{n} \sum_i (t^{(i)} - mx^{(i)} - c) = 0$$

$$-\sum_i t^{(i)} + m \sum_i x^{(i)} + \sum_i c = 0$$

$$m \sum_i x^{(i)} + c \sum_i 1 = \sum_i t^{(i)}$$

$$\begin{bmatrix} \sum_i x^{(i)} & \sum_i 1 \end{bmatrix} \begin{bmatrix} m \\ c \end{bmatrix} = \begin{bmatrix} \sum_i t^{(i)} \end{bmatrix}$$

$$1 \times 2$$

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Line Fitting: Least Square Error Solutions

$$\begin{bmatrix} \sum_i (x^{(i)})^2 & \sum_i x^{(i)} \end{bmatrix} \begin{bmatrix} m \\ c \end{bmatrix} = \begin{bmatrix} \sum_i t^{(i)} x^{(i)} \end{bmatrix} \quad \Bigg| \quad \begin{bmatrix} m \sum_i x^{(i)} & c \sum_i 1 \end{bmatrix} \begin{bmatrix} m \\ c \end{bmatrix} = \begin{bmatrix} \sum_i t^{(i)} \end{bmatrix}$$

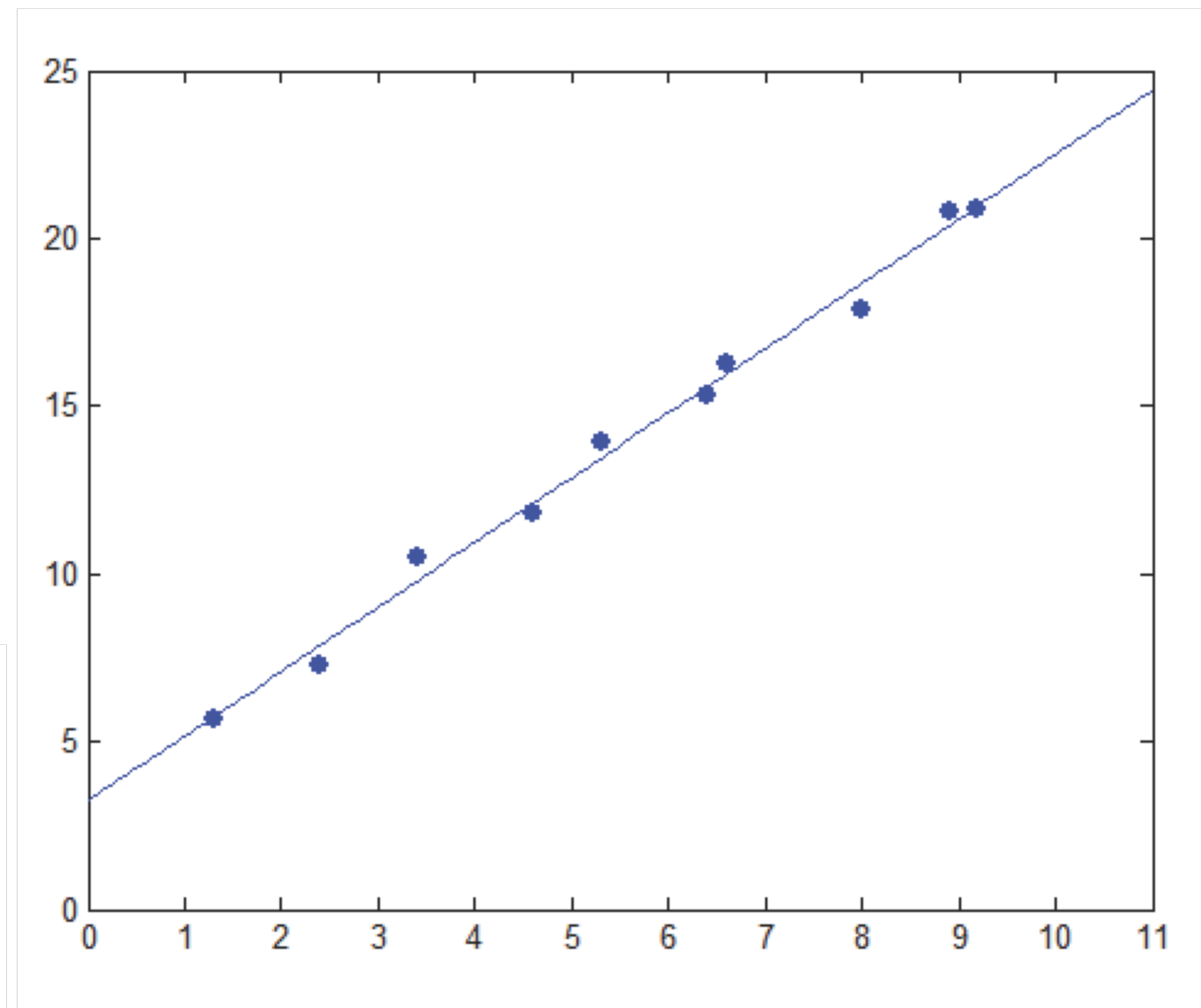
$$\begin{bmatrix} \sum_i (x^{(i)})^2 & \sum_i x^{(i)} \\ \sum_i x^{(i)} & \sum_i 1 \end{bmatrix} \begin{bmatrix} m \\ c \end{bmatrix} = \begin{bmatrix} \sum_i t^{(i)} x^{(i)} \\ \sum_i t^{(i)} \end{bmatrix}$$

$2 \times 2 \qquad \qquad 2 \times 1 \qquad \qquad 2 \times 1$

$$\mathbf{Ax} = \mathbf{b}$$

$$\begin{bmatrix} 380.63 & 56.1 \\ 56.1 & 10 \end{bmatrix} \begin{bmatrix} m \\ c \end{bmatrix} = \begin{bmatrix} 914.68 \\ 140.4 \end{bmatrix}$$

Solution: $m = 1.9274$ $c = 3.227$



Linear Fitting: Least Square Error Solutions

Line only

$$y = mx + c$$

General Solution

$$\mathbf{w} = \{w_0, w_1, w_2, \dots\}^T$$

$$\mathbf{x} = \{1, x_1, x_2, \dots\}^T$$

$$y^{(i)} = \mathbf{w}^T \mathbf{x}^{(i)} = \sum_j w_j x_j^{(i)}$$

Linear Fitting: Least Square Error Solutions

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$$y = mx + c$$

$$E = \sum_i (t^{(i)} - mx^{(i)} - c)^2$$

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$$E = \sum_i (t^{(i)} - y^{(i)})^2 = \sum_i (t^{(i)} - \mathbf{w}^T \mathbf{x}^{(i)})^2$$

Linear Fitting: Least Square Error Solutions

Line only

$$y = mx + c$$

$$E = \sum_i (t^{(i)} - mx^{(i)} - c)^2$$

$$\frac{\partial E}{\partial m} = -\frac{2}{n} \sum_i (t^{(i)} - mx^{(i)} - c)x^{(i)}$$

$$\frac{\partial E}{\partial c} = -\frac{2}{n} \sum_i (t^{(i)} - mx^{(i)} - c)$$

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$$\frac{\partial E}{\partial w_j} = -\frac{2}{n} \sum_i (t^{(i)} - y^{(i)})x_j^{(i)}$$

Linear Fitting: Least Square Error Solutions

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$$y = mx + c$$

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$$\frac{\partial E}{\partial c} = -\frac{2}{n} \sum_i (t^{(i)} - mx^{(i)} - c)$$

~~$$\begin{bmatrix} \sum_i (x^{(i)})^2 & \sum_i x^{(i)} \\ \sum_i x^{(i)} & \sum_i 1 \end{bmatrix} \begin{bmatrix} m \\ c \end{bmatrix} = \begin{bmatrix} \sum_i t^{(i)} x^{(i)} \\ \sum_i t^{(i)} \end{bmatrix}$$~~

General Solution

$$\mathbf{w} = \{w_0, w_1, w_2, \dots\}^T$$

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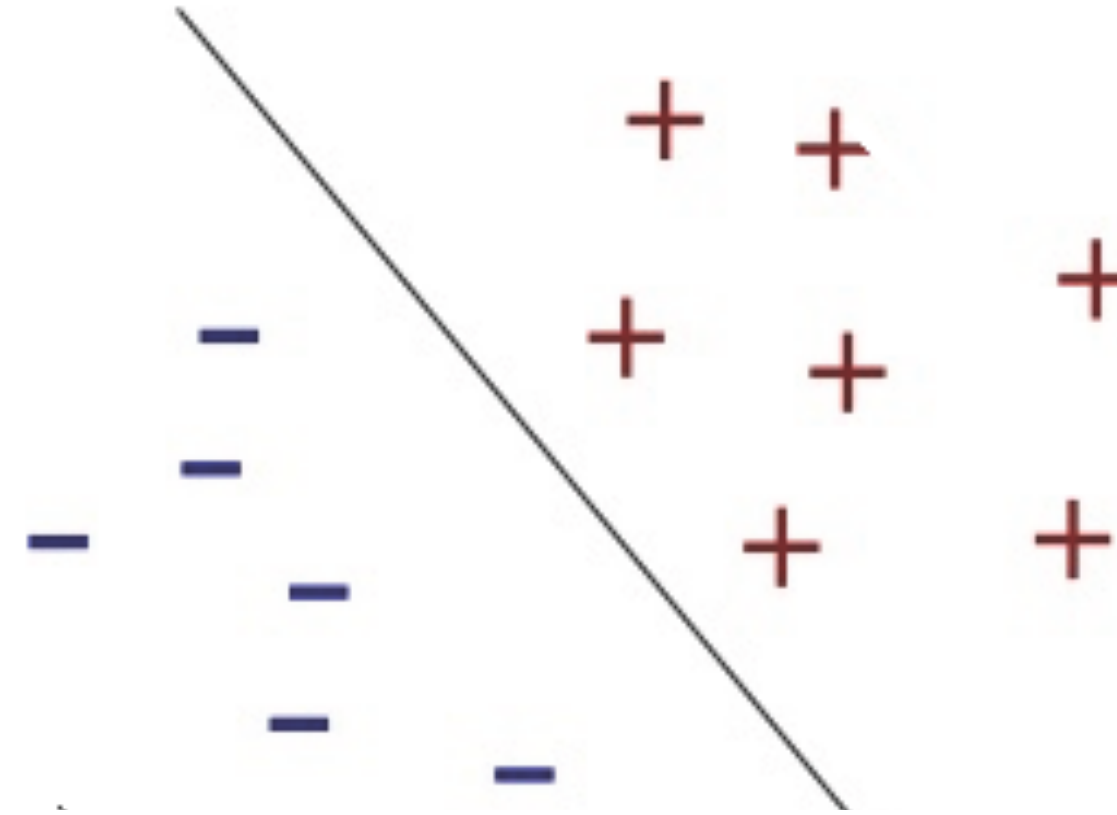
Linear Classifier

$$\mathbf{w} = \{w_0, w_1, w_2, \dots\}^T$$

$$\mathbf{x} = \{1, x_1, x_2, \dots\}^T$$

$$y^{(i)} = \mathbf{w}^T \mathbf{x}^{(i)} = \sum_j w_j x_j^{(i)}$$

$$E = \sum_i (t^{(i)} - y^{(i)})^2 = \sum_i (t^{(i)} - \mathbf{w}^T \mathbf{x}^{(i)})^2$$



Linear Classifier

$$\mathbf{w} = \{w_0, w_1, w_2, \dots\}^T$$

$$\mathbf{x} = \{1, x_1, x_2, \dots\}^T$$

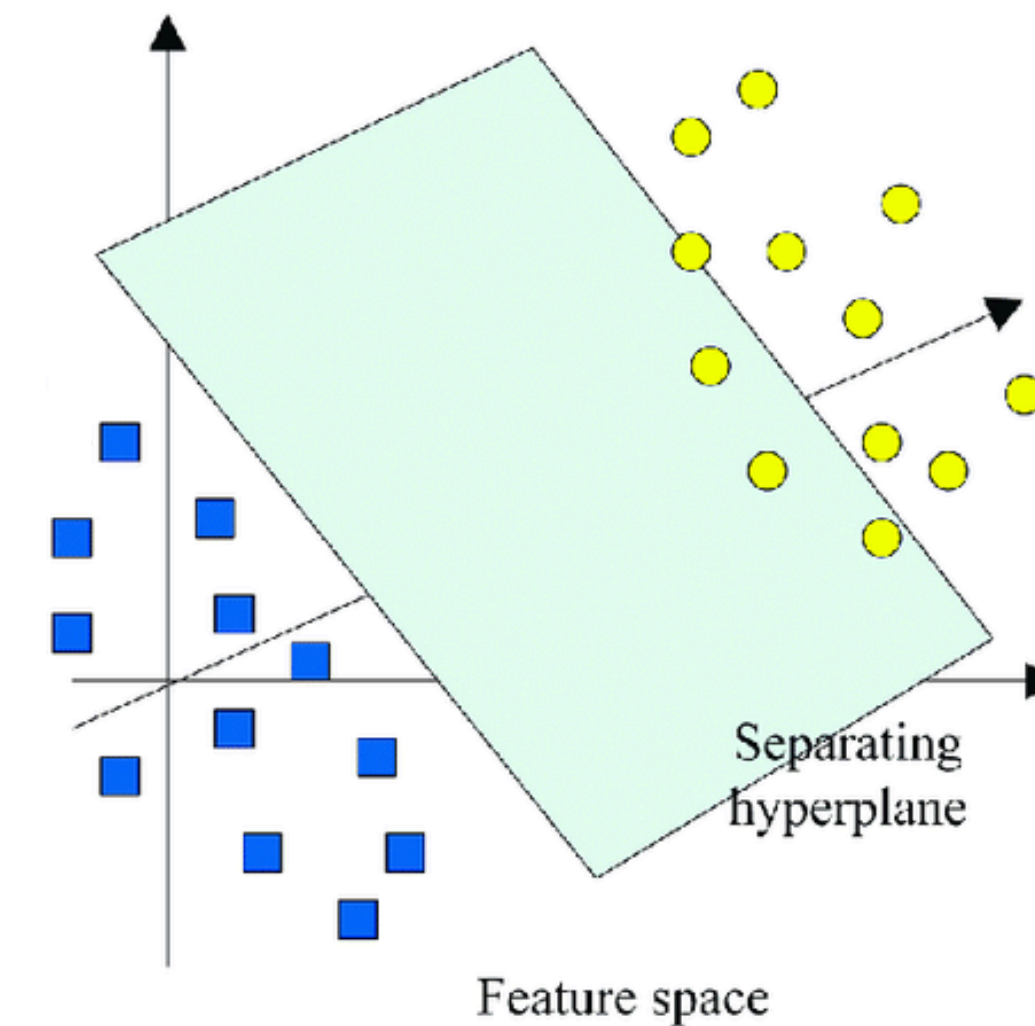
$$y^{(i)} = \mathbf{w}^T \mathbf{x}^{(i)} = \sum_j w_j x_j^{(i)}$$

$$t^{(i)} \in \{-1, +1\}$$

$$E = \sum_i (t^{(i)} - y^{(i)})^2 = \sum_i (t^{(i)} - \mathbf{w}^T \mathbf{x}^{(i)})^2$$

$$\frac{\delta E}{\delta w_j} = - \sum_i (t^{(i)} - y^{(i)}) x_j^{(i)}$$

$$\begin{cases} +1 & y > 0 \\ -1 & y \leq 0 \end{cases}$$



$$y = mc + b$$

$$Ax + By + Cz + D = 0$$

$$z = (-Ax - By - D)/C$$

$$w_0 = -A/C$$

$$w_1 = -B/C$$

$$w_2 = -D/C$$

$$2 \times 10 - 1 \times 11 - 1 \times 12 = -3$$

$$2 \times 10 - 1 \times 14 - 1 \times 9 = -3$$

Lbs	Width	Height	y
1	3	2	Insect
1	4	3	Insect
2	11	12	Bird
2	14	9	Bird

$$\triangleright \mathbf{w} = \{10, -1, -1\}$$

$$\triangleright 1 \times 10 + (-1) \times 3 + (-1) \times 2 > 0$$